A FLAT MONOMORPHISM OF ALGEBRAIC SPACES IS REPRESENTABLE BY SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 286-289 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 291-313, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Flat monomorphisms of algebraic spaces, representability by schemes, and the local structure of a flat monomorphism; the reduction is retained.

Source reference. Original lemma decent-spaces-lemma-generalizations-lift-flat: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 313–313. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-descent-lemma-descending-property-isomorphism: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 313–313. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-limits-lemma-enough-local: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 313–313. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-subscheme: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 313–313. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (David Rydh)**.** *A flat monomorphism of algebraic spaces is representable by schemes.*

Proof. Let $f : X \to Y$ be a flat monomorphism of algebraic spaces. To prove f is representable, we have to show $X \times_Y V$ is a scheme for every scheme V mapping to

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Y. Since being a scheme is local (Properties of Spaces, Lemma spaces-properties-lemma-subscheme), we may assume V is affine. Thus we may assume $Y = \mathrm{Spec}(B)$ is an affine scheme. Next, we can assume that X is quasi-compact by replacing X by a quasi-compact open. The space X is separated as $X \to X \times_{\mathrm{Spec}(B)} X$ is an isomorphism. Applying Limits of Spaces, Lemma spaces-limits-lemma-enough-local we reduce to the case where B is local, $X \to \mathrm{Spec}(B)$ is a flat monomorphism, and there exists a point $x \in X$ mapping to the closed point of $\mathrm{Spec}(B)$. Then $X \to \mathrm{Spec}(B)$ is surjective as generalizations lift along flat morphisms of separated algebraic spaces, see Decent Spaces, Lemma decent-spaces-lemma-generalizations-lift-flat. Hence we see that $\{X \to \mathrm{Spec}(B)\}$ is an fpqc cover. Then $X \to \mathrm{Spec}(B)$ is a morphism which becomes an isomorphism after base change by $X \to \mathrm{Spec}(B)$. Hence it is an isomorphism by fpqc descent, see Descent on Spaces, Lemma spaces-descent-lemma-descending-property-isomorphism. $\square$